

Economic Convergence Analysis using Bootstrap Methods

AppStats Project Report

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Abstract. This project tests whether poor countries in 2000 grew faster than rich countries over 2000–2020, a pattern known as beta-convergence (Barro & Sala-i-Martin, 1992). We use GDP per capita data from Gapminder for 193 countries and split them into two groups based on the global median income in 2000. We then compare their median log-growth rates using a non-parametric bootstrap ($B = 2000$) to measure uncertainty. The observed difference is $\hat{\theta} = 0.185$ (SE = 0.053), with a 95% confidence interval of [0.081, 0.290]. Since this interval does not include zero, the result supports the convergence hypothesis.

1 Introduction

The beta-convergence hypothesis says that poor countries tend to grow faster than rich ones. The idea comes from basic economics: when a country has little capital, each new investment brings a high return. As a country gets richer, returns on investment go down. This means poor countries should grow faster and, over time, catch up with richer ones. This theory was studied by Barro and Sala-i-Martin (1992).

Our question: Did countries that were below the global median income in 2000 grow faster than richer countries between 2000 and 2020?

2 Data and Variables

We use GDP per capita from the Gapminder Data Portal. The values are adjusted for purchasing power parity (PPP), which means the numbers account for differences in the cost of living between countries. All values are in constant 2021 international dollars, so inflation does not affect the comparison.

The dataset has 193 countries with complete data for both 2000 and 2020. We split them into two groups using the global median GDP per capita in 2000 (\$9,351):

- **Low-income group:** 96 countries below the median.
- **High-income group:** 97 countries at or above the median.

A more official split would use the World Bank income threshold (around \$12,000 in 2000), which gives roughly 35 rich countries and 158 poor countries. However, such an unequal split is not ideal for bootstrap: with only 35 countries in one group, the empirical distribution approximates the true distribution less well, the median can only take a small number of distinct values across replications, and the confidence interval becomes less stable. The median split produces balanced groups of similar size, which gives more reliable bootstrap estimates in both groups.

For each country i , we compute the log-growth rate over 20 years:

$$g_i = \ln\left(\frac{\text{GDP}_{i,2020}}{\text{GDP}_{i,2000}}\right) \quad (1)$$

We use the log-ratio instead of a simple percentage change because it is symmetric: a country that doubles its GDP and a country that halves it get values of equal size with opposite signs.

3 Estimation and Bootstrap

Statistic of interest

We use the median instead of the mean to compare the two groups, because some countries have very extreme growth values (very high or very low) that would distort the mean. Our statistic is:

$$\hat{\theta} = \text{median}(g_i \mid \text{poor}) - \text{median}(g_i \mid \text{rich}) \quad (2)$$

Why bootstrap?

Computing $\hat{\theta}$ once is not enough. We need to know how uncertain this value is, and in particular, whether zero is inside the 95% confidence interval. If it is, the difference we observe could just be random, and we cannot conclude anything about convergence.

The problem is that the median has no simple formula for its standard error. We therefore use the non-parametric bootstrap to approximate the distribution of $\hat{\theta}$. We use $B = 2000$ replications, which is enough to get stable estimates of the confidence interval bounds.

Algorithm

For $b = 1, \dots, B = 2000$:

1. Draw a sample of size $n_{\text{poor}} = 96$ **with replacement** from the low-income group.
2. Draw a sample of size $n_{\text{rich}} = 97$ **with replacement** from the high-income group.
3. Compute: $\hat{\theta}^{*b} = \text{median}(\mathbf{g}_{\text{poor}}^{*b}) - \text{median}(\mathbf{g}_{\text{rich}}^{*b})$

We resample *within each group separately*. If we re-sampled from the full dataset before the split, the median threshold would change at each step, moving countries between groups. This would break the analysis, because we would no longer be measuring the same comparison each time.

The standard error and confidence interval are then:

$$\widehat{SE}_B = \sqrt{\frac{1}{B-1} \sum_{b=1}^B (\hat{\theta}^{*b} - \bar{\theta}^*)^2} \quad (3)$$

$$CI_{95\%} = [\hat{\theta}^{*(0.025)}, \hat{\theta}^{*(0.975)}] \quad (4)$$

4 Results

The median log-growth rate is 0.410 for low-income countries and 0.225 for high-income countries, giving $\hat{\theta} = 0.185$. Figure 1 shows that the distributions are right-skewed, which confirms why we use the median. The bootstrap gives a standard error of 0.053 and a 95% confidence interval of [0.081, 0.290] (Figure 2).

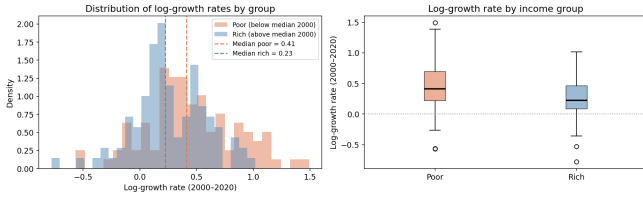


Figure 1: Log-growth rate distributions by group. Dashed lines show the group medians.

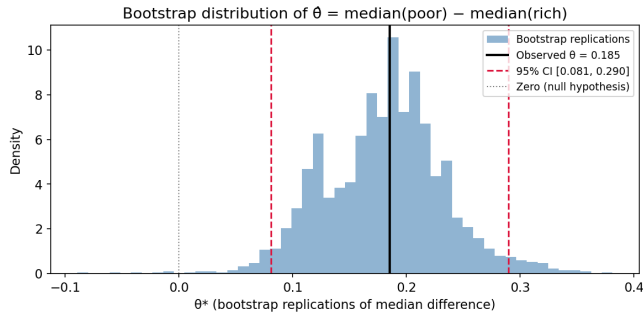


Figure 2: Bootstrap distribution of $\hat{\theta}$ ($B = 2000$). Red dashed lines: 95% CI bounds. Grey dotted line: zero.

5 Interpretation

The 95% confidence interval [0.081, 0.290] does not include zero. This means we can reject the idea that the two groups grew at the same rate. Low-income countries in 2000 grew significantly faster than high-income countries over 2000–2020, which is consistent with the beta-convergence hypothesis.

However, this is a statistical *association*, not a causal effect. Our analysis only separates countries by their

income in 2000. But poor and rich countries are different in many other ways too, for example their institutions, education levels, or natural resources. All of these factors also affect growth. So we cannot say that being poor *caused* faster growth. To show that, we would need to control for all these other differences, for example using a multivariate regression. This is outside the scope of this project.

Finally, the bootstrap relies on the empirical distribution $\hat{F}_n$ approximating the true distribution F , which requires a large n . With 96 and 97 countries per group, this approximation is reasonable in practice, but not perfect. The median is particularly sensitive to this: with only 96 data points, the bootstrap median can only take a limited number of distinct values, making the confidence interval bounds less precise than they would be with a larger sample. This is a natural limit of the data: there are only 193 countries in the world.